

Project Progress Dashboard

Digital Newborn Hearing Screening Management Platform – replacing paper-based UNHS workflow

2/6

PHASES ACTIVE

8

TASKS DONE

3

IN PROGRESS

14

TASKS REMAINING

Overall Completion

32%

ROADMAP – ALL PHASES

PHASE 01

COMPLETE

Environment Setup

Week 1

- VSCode + Flutter setup
- FastAPI + Python environment
- PostgreSQL local setup
- Git + GitHub repo

PHASE 02

IN PROGRESS

Database + Auth

Week 1-2

- DB schema design
- JWT + RBAC (4 roles)
- Protected routes + dependencies
- Test users seeded
- PostgreSQL connection – verify
- Encryption setup (AES-256, TLS 1.3)
- Immutable audit log

PHASE 03

UPCOMING

Backend API

Week 2-3

- Screening API (CRUD, validation)
- Patient records API (anon baby IDs)

PHASE 04

UPCOMING

Flutter Mobile App

Week 3-6

- Project structure + routing
- Auth screens + role-based routing

- Follow-up queue API
- MoH report API (Excel, KKM format)
- API docs (OpenAPI / Swagger)

- Screener module (<60s flow)
- QR scanning (baby wristband)
- Offline sync (SQLite)
- Bilingual BM/EN (i18n)

PHASE 05

UPCOMING

Web Portal + Cloud

Week 6–9

- Coordinator web dashboard (React)
- KKM reporting dashboard
- AWS Malaysia setup (ap-southeast-1)
- Docker + CI/CD deployment

PHASE 06

UPCOMING

Testing + Pilot Prep

Week 9–12

- Backend tests (pytest, API)
- Flutter device testing (Android)
- PDPA compliance checklist
- APK build + distribution
- Monitoring + logs (CloudWatch)
- User training guide

AUTH SYSTEM — FILES BUILT

.py

auth/auth.py

Password hashing +
JWT
generation/validation

.py

auth/dependencies.py

Role-based route
dependencies

.py

auth/models.py

Pydantic schemas
for auth

.py

routers/auth_router.py

Login endpoint +
protected routes

.py

db/database.py

SQLAlchemy engine +
session setup

.py

seed.py

Test data seeder
script

.md

AUTH_SETUP.md

Comprehensive auth
setup guide

.md

ARCHITECTURE.md

System flow diagrams

RBAC — TEST USERS

4 ROLES IMPLEMENTED

screener

screener@test.com

coordinator

coordinator@test.com

hospital_admin

admin@test.com

SECURITY STACK

bcrypt

password hashing

python-jose

JWT tokens

HTTPBearer

security scheme

moh

moh@test.com

dependency injection

route protection

BLOCKERS + NEXT STEPS

BLOCKER PostgreSQL connection needs verification. `dtdatabase.py` runs without error but live DB connectivity has not been confirmed end-to-end through the API layer.

IMMEDIATE NEXT STEPS — PHASE 2 COMPLETION

- | | | |
|--|---|---|
| 01 Verify PostgreSQL connection through FastAPI startup | 02 Implement AES-256 encryption at rest + TLS 1.3 in transit | 03 Build append-only timestamped audit log table |
|--|---|---|